



AI MOBILE APP BOOTCAMP CURRICULUM



Month 1: Coding Foundations, GitHub & AI Tools

Week 1: Introduction to Software Development & AI

- What is software? What is AI?
- Real-world examples of AI in apps (YouTube, TikTok, Netflix)
- Setting up tools: VS Code, GitHub, GitHub Copilot, ChatGPT
- First interaction: Use ChatGPT for logic building and code suggestions

Week 2: JavaScript Basics

- Variables, data types, functions
- Conditionals, loops, and arrays
- Practice: Build a math quiz game using JavaScript

Week 3: GitHub & GitHub Copilot

- What is Git and GitHub? Why use version control?
- Creating and managing repositories
- Using GitHub Copilot to write and understand code

Week 4: Problem Solving with ChatGPT

- Prompting strategies for coding help and productivity
- Debugging and explanation practice with ChatGPT
- Mini challenge: Complete small projects using ChatGPT + Copilot



Month 2: Mobile App Development with React Native & Expo

Week 5: React Native Basics

- What is React Native? Why Expo?
- Installing Expo CLI and using Expo Go
- Building your first screen (text, image, buttons)

Week 6: Components, Props & State

- Functional components and passing data (props)
- Using state for dynamic content
- Practice: Build a personal profile or bio app

Week 7: Multi-Screen Apps & Navigation

- Setting up navigation with React Navigation
- Creating and linking multiple screens
- Practice: Build a mobile story app or interactive quiz

Week 8: Mini App Project

- Choose an app idea (task tracker, quote generator, etc.)
- Build UI, link screens, and submit on GitHub
- Share project and get peer feedback



Month 3: AI Concepts + Python Model Building

Week 9: Introduction to AI & Regression Algorithm

- What is machine learning? What is regression?
- How apps use prediction (e.g., exam score predictor, budget tracker)
- Build your first regression model in Python using scikit-learn

Week 10: Random Forest Algorithm

- What are decision trees and random forest?
- Training and testing with example datasets
- Practice: Predict performance using student data

Week 11: Recommendation Engines

- How apps suggest what users like (movies, books)
- Build a simple recommender system using Python
- Practice: Recommend content based on user preferences

Week 12: Preparing Models for Deployment

- Save models using `joblib`
- Understand JSON and how data flows from app to model
- Simulate model responses in Python



Month 4: Flask API, Deployment & Final Projects

Week 13: Flask for Serving AI Models

- What is an API? What does Flask do?
- Build API endpoints to serve predictions
- Test your API with Postman

Week 14: Deploying Flask on AWS

- Set up an AWS EC2 instance
- Host your Flask app so it's accessible online
- Share your API endpoint

Week 15: Connecting AI to React Native

- Use `fetch` in React Native to call your Flask API
- Display predictions or recommendations in your mobile app
- Practice: Update your earlier project with live AI integration

Week 16: Final Project Week

- Plan, build, and deploy a complete AI-powered mobile app
- Use GitHub Copilot and ChatGPT for support
- Present your app: explain the problem it solves, show features, and describe how the AI works

Final Skillset

- **Languages:** JavaScript (React Native), Python (AI)
- **Mobile Dev:** React Native with Expo
- **AI Models:** Regression, Random Forest, Recommendation Engine
- **APIs:** Flask for serving models
- **Deployment:** AWS EC2
- **Tools:** GitHub, GitHub Copilot, ChatGPT